

GEUVADIS Gene Expression GREML Workflow Handoff

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This document is a knowledge-transfer guide for the GEUVADIS gene expression heritability workflow in this repository. It is written for the next person who needs to understand, rerun, audit, or extend the analysis after the original workflow owner leaves the project.

The workflow estimates gene-level SNP heritability of RNA expression phenotypes using GEUVADIS RNA-seq data, 1000 Genomes genotype data, GTEx-inspired expression processing, and GCTA GREML. The repository also contains downstream diagnostic analyses to compare expression choices, normalization choices, SNP sets, and expression distribution properties.

1. Scientific Question

The central question is:

How much of gene expression variation across European GEUVADIS individuals can be explained by genotype-derived relatedness, and how sensitive are those heritability estimates to expression quantification, normalization, SNP set, and expression distribution shape?

Operationally, each gene is treated as a quantitative phenotype. GCTA GREML is used to estimate the fraction of phenotypic variance explained by a genomic relationship matrix:

$$h^2_{\text{GREML}} = V(G) / V_p$$

where 'V(G)' is the additive genetic variance component estimated from the GRM and 'Vp' is total phenotypic variance after the model accounts for covariates.

The workflow asks this question under multiple preprocessing choices:

- * SNP set: all available SNPs in the genotype prefix versus HapMap3/no-MHC SNPs.
- * Expression source: TPM-derived expression versus TMM-normalized count-derived expression.
- * Phenotype normalization: raw expression scale versus inverse-normal transformed expression.
- * Covariates: genotype PCs and PEER factors.

The broader scientific motivation is that expression heritability estimates can be sensitive to the phenotype scale. Low expression, zero inflation, skewed expression distributions, and different normalization choices can all change how GREML partitions variance.

2. Previous Research And What This Workflow Mimics

GEUVADIS and 1000 Genomes

GEUVADIS generated RNA-seq data for lymphoblastoid cell lines from 1000 Genomes individuals. The public GEUVADIS collection contains mRNA and small RNA sequencing from over 460 samples in CEU, FIN, GBR, TSI, and YRI. This workflow focuses on the European-ancestry groups used in many GEUVADIS analyses:

- * British: GBR
- * Finnish: FIN
- * Tuscan: TSI
- * Utah residents with Northern and Western European ancestry: CEU

The project uses GEUVADIS expression data because it has both population-scale RNA-seq and matched 1000 Genomes genotype data. That makes it a good test case for expression heritability: for the same individuals, we can construct gene expression phenotypes and genotype-derived GRMs.

Hernandez et al.-style expression heritability framing

Hernandez et al. studied gene expression heritability in GEUVADIS and emphasized how variant frequency classes, rare variants, and expression architecture affect estimated cis heritability. In their GEUVADIS European analysis, they started from European individuals and removed cryptically related samples using 1000 Genomes relatedness/identity-by-state information, ending with an unrelated EUR set.

This workflow mimics the Hernandez-style sample logic by:

- * Starting from GEUVADIS EUR metadata.
- * Mapping RNA-seq run IDs to 1000 Genomes genotype IDs.
- * Optionally intersecting the mapped EUR sample set with 'unrelated_keep_file'.
- * Using that sample set consistently for GRM, PCA, phenotype prep, and GREML.

Important distinction:

```
Hernandez-style pruning = external 1000G unrelated sample keep list  
not necessarily GCTA --grm-cutoff 0.025
```

The workflow now exposes this as:

```
params.unrelated_keep_file
```

GTEx-style expression phenotype processing

The workflow intentionally follows GTEx-like expression processing choices:

- * Keep genes with TPM ≥ 0.1 in at least 20% of samples.
- * Keep genes with counts ≥ 6 in at least 20% of samples.
- * Use TMM normalization for counts.
- * Use inverse-normal transformation per gene for normalized expression phenotypes.
- * Use genotype PCs and PEER factors as covariates.
- * Use a sample-size-based rule for PEER factors:
'15, 30, 45, 60' for increasing sample-size bins.

GTEx used these types of steps to make expression phenotypes more robust for linear modeling and eQTL mapping. This workflow reuses that logic before GREML.

TMM normalization rationale

TMM, the trimmed mean of M-values method from Robinson and Oshlack, was developed because simple library-size scaling can be biased when RNA composition differs between samples. If a sample has a few highly expressed genes, those genes take up sequencing "real estate" and can make all other genes appear lower by simple proportional scaling. TMM estimates sample-specific scaling factors after trimming extreme log-fold changes and intensity values, producing effective library sizes for cross-sample comparison.

In this pipeline, when 'expression_source=tmm', counts are converted to TMM-normalized CPM:

```
DGEList(counts)  
calcNormFactors(method = "TMM")  
cpm(normalized.lib.sizes = TRUE, log = FALSE, prior.count = 0)
```

GCTA GREML rationale

GCTA GREML uses a genomic relationship matrix as the covariance structure for random additive genetic effects. If genetically more similar individuals also have more similar expression for a gene, GREML attributes some expression variance to genotype. The output 'V(G)/Vp' is interpreted as the SNP heritability captured by the SNPs used to build the GRM.

In this workflow, the GRM is built once per run setting and the same GRM is used across genes. Each gene is then selected from a multi-phenotype file using GCTA's '--mpheno' option.

3. Repository Layout

Primary files:

```
README.md
run_greml.sh
nf/main.nf
nf/nextflow.config
nf/bin/prepare_phenotypes.R
scripts/
docs/
```

Important documentation already in the repo:

```
docs/how_to_run_greml_nextflow.md
docs/how_to_prepare_gtex_like_phenotypes.md
docs/how_to_run_downstream_analysis.md
docs/downstream_analysis_methods.md
```

Important analysis scripts:

```
scripts/plot_greml_h2_summary_boxplot.R
scripts/pairwise_h2_comparisons.R
scripts/tmm_expression_h2_deciles.R
scripts/tmm_raw_skewness_analysis.R
scripts/tpm_tmm_peer_correlations.R
scripts/downstream_h2_regulatory_repeat_analysis.R
```

4. Main HPC Paths

Project root on HPC:

```
/gpfs/data/mostafavilab/sool/analysis/GeneExpression/20260428_GE_GEUVDIS_v2/GeneExpression
```

Run directory root:

```
/gpfs/data/mostafavilab/sool/analysis/GeneExpression/20260428_GE_GEUVDIS_v2/GeneExpression/runs
```

Nextflow scratch/work root:

```
/gpfs/scratch/sl8085/nextflow_work/greml_production
```

RNA-seq matrices from the previous Salmon workflow:

```
/gpfs/data/mostafavilab/sool/analysis/GeneExpression/20260125_salmon_nextflow/results/matrices/gene_tpm.
tsv
/gpfs/data/mostafavilab/sool/analysis/GeneExpression/20260125_salmon_nextflow/results/matrices/gene_coun
ts.tsv
```

Default genotype prefix:

```
/gpfs/data/mostafavilab/shared_data/LDSC_data/1000G_EUR_Phase3_plink/1000G_EUR.QC.22
```

This resolves to:

```
1000G_EUR.QC.22.bed
1000G_EUR.QC.22.bim
1000G_EUR.QC.22.fam
```

The current prefix ends with '.22'. A successor should verify whether this is intentionally chromosome 22 only or whether the intended production GRM should be built from genome-wide merged chromosomes. This is one of the most important handoff checks.

GCTA executable:

```
/gpfs/data/mostafavilab/programs/GCTA/gcta-1.95.0-linux-kernel-3-x86_64/squashfs-root/AppRun
```

HM3/no-MHC SNP list:

```
/gpfs/data/mostafavilab/shared_data/LDSC_data/hm3_no_mhc_snps.txt
```

GEUVADIS SDRF metadata:

```
https://www.ebi.ac.uk/arrayexpress/files/E-GEUV-1/E-GEUV-1.sdrf.txt
```

Detailed input file inventory

This section is intentionally logistics-heavy. The goal is that the next person can identify each input on HPC, understand what biological or computational information it carries, and know why the workflow needs it.

Project repository root:

```
/gpfs/data/mostafavilab/sool/analysis/GeneExpression/20260428_GE_GEUVDIS_v2/GeneExpression
```

What it contains:

- * 'run_greml.sh', the Slurm-facing driver script.
- * 'nf/main.nf', the Nextflow workflow definition.
- * 'nf/nextflow.config', the default file paths, parameters, resources, and Slurm profile.
- * 'nf/bin/prepare_phenotypes.R', the R script that turns Salmon matrices plus metadata/covariates into GCTA-compatible phenotype and covariate files.
- * 'scripts/', downstream plotting and QC scripts.
- * 'docs/', handoff and methods documentation.

Why we use it:

- * This is the reproducible analysis root. Submitting from this directory lets 'run_greml.sh' resolve the intended Nextflow project files instead of accidentally using a Slurm spool path.
- * The driver script checks the SHA-256 and modification time of 'nf/bin/prepare_phenotypes.R' before launching. That protects against silently running a stale phenotype-preparation script.

Upstream GEUVADIS FASTQ input used by the Salmon workflow:

```
/gpfs/data/mostafavilab/reference_files/RNAseq/GEUVADIS/ERR*.fastq.gz
```

What it contains:

- * GEUVADIS RNA-seq reads, one FASTQ per ENA run ID such as 'ERR...'.
* These are the raw sequencing inputs from which transcript and gene expression were quantified.

Why we use it:

- * GEUVADIS provides RNA-seq from lymphoblastoid cell lines with matched 1000 Genomes genotypes.
- * The ENA run IDs are later linked to individual IDs through the SDRF metadata, making it possible to match expression phenotypes to genotype samples.

GRCh37/hg19 transcriptome annotation used for Salmon quantification:

```
/gpfs/data/mostafavilab/reference_files/transcriptome/hg19/Homo_sapiens.GRCh37.75.gtf.gz
```

What it contains:

- * Ensembl GRCh37 release 75 gene and transcript annotations.
- * The upstream Salmon workflow parses 'transcript_id' and 'gene_id' from this GTF to create 'tx2gene.tsv'.

Why we use it:

- * GEUVADIS and 1000 Genomes-era resources are commonly aligned to GRCh37/hg19.
- * Using a matching GRCh37 transcriptome avoids mixing genome builds between RNA quantification and 1000 Genomes genotype resources.
- * The GTF is required to aggregate transcript-level Salmon estimates to gene-level estimates.

GRCh37/hg19 cDNA fasta used for Salmon indexing:

```
/gpfs/data/mostafavilab/reference_files/transcriptome/hg19/Homo_sapiens.GRCh37.75.cdna.all.fa.gz
```

What it contains:

- * Ensembl GRCh37 release 75 transcript sequences.
- * Salmon uses this fasta to build the transcriptome index.

Why we use it:

- * Salmon quantifies reads against transcript sequences. The cDNA fasta is the reference sequence input for the quasi-mapping index.
- * It is paired with the GTF above, so transcript IDs in the fasta can be mapped back to gene IDs consistently.

Salmon TPM matrix:

```
/gpfs/data/mostafavilab/sool/analysis/GeneExpression/20260125_salmon_nextflow/results/matrices/gene_tpm.tsv
```

What it contains:

- * A gene-by-sample matrix produced by the previous Salmon workflow.
- * First column is the gene ID; remaining columns are GEUVADIS run/sample columns.
- * Values are Salmon gene-level TPM estimates.

Where it enters this workflow:

- * 'nf/nextflow.config' sets 'params.tpm_file' to this path.
- * 'PREPARE_PHENOTYPES' passes it to 'nf/bin/prepare_phenotypes.R'.
- * The R script reads it with 'fread()', identifies the gene column, strips Ensembl version suffixes if present, maps sample columns to GEUVADIS run IDs, and applies the GTEx-style expression filter.

Why we use it:

- * TPM normalizes within a sample for transcript/gene length and sequencing depth. It is useful for comparing relative expression abundance across genes within a sample.
- * TPM is part of the expression filter even when the phenotype source is TMM. A gene must have TPM above threshold in enough samples and count support in enough samples.
- * TPM can also be used directly as the phenotype source when '--expression_source tpm'.

Scientific caveat:

- * TPM is compositional. If a few genes dominate a sample, TPM values for other genes can shift even when absolute molecule counts do not. This is one reason the workflow compares TPM-derived phenotypes to TMM-derived phenotypes.

Salmon estimated-count matrix:

```
/gpfs/data/mostafavilab/sool/analysis/GeneExpression/20260125_salmon_nextflow/results/matrices/gene_counts.tsv
```

What it contains:

- * A gene-by-sample matrix from Salmon 'NumReads', aggregated to gene level.
- * First column is gene ID; remaining columns correspond to GEUVADIS run/sample columns.
- * Values are estimated counts, not necessarily integers, because Salmon probabilistically assigns ambiguous reads.

Where it enters this workflow:

- * 'nf/nextflow.config' sets 'params.counts_file' to this path.
- * 'PREPARE_PHENOTYPES' passes it to 'prepare_phenotypes.R'.
- * The R script uses counts for the GTEx-style count filter and for TMM normalization when '--expression_source tmm'.

Why we use it:

- * TMM is defined on count data. It estimates sample-specific normalization factors that correct for library-size and composition effects.
- * Counts preserve information about read support. A gene with TPM above threshold but very low count support can be unstable for heritability estimation, so both TPM and count filters are required.

GEUVADIS SDRF metadata:

```
https://www.ebi.ac.uk/arrayexpress/files/E-GEUV-1/E-GEUV-1.sdrf.txt
```

What it contains:

- * Sample-level GEUVADIS metadata from ArrayExpress.
- * Columns include RNA-seq run identifiers such as 'ENA_RUN', ancestry/population labels, and individual/sample identifiers.

Where it enters this workflow:

- * 'FILTER_EUROPEANS' downloads it to 'sdrf.txt' inside the Nextflow task work directory.
- * The process detects the ancestry column and the 'ENA_RUN' column.
- * It keeps the configured European ancestry labels and writes 'eur_map.tsv', which maps RNA run IDs to genotype FID/IID.

Why we use it:

- * The expression matrices are indexed by RNA-seq run/sample columns, while the genotype files are indexed by 1000 Genomes FID/IID.
- * The SDRF is the bridge between those two naming systems.
- * It also allows the workflow to restrict to European GEUVADIS populations, reducing ancestry-driven structure in expression and genotype covariance.

European ancestry labels used from the SDRF:

```
British, Finnish, Tuscan, Utah
```

What they represent:

- * British corresponds to GBR.
- * Finnish corresponds to FIN.
- * Tuscan corresponds to TSI.
- * Utah corresponds to CEU.

Why we use them:

- * These are the GEUVADIS European groups that match the 1000 Genomes EUR genotype resource.
- * Restricting ancestry reduces confounding from major continental population structure. Residual European substructure is handled with genotype PCs.

1000 Genomes EUR PLINK prefix:

```
/gpfs/data/mostafavilab/shared_data/LDSC_data/1000G_EUR_Phase3_plink/1000G.EUR.QC.22
```

Files implied by the prefix:

```
1000G.EUR.QC.22.bed  
1000G.EUR.QC.22.bim  
1000G.EUR.QC.22.fam
```

What '.bed' contains:

- * Binary PLINK genotype matrix.
- * Rows/variants and individuals are encoded compactly for efficient genotype operations.

What '.bim' contains:

- * Variant metadata: chromosome, SNP ID, genetic position, base-pair position, and alleles.
- * GCTA uses this to know which variants are included and to apply optional SNP extraction.

What '.fam' contains:

- * Sample metadata: family ID, individual ID, father ID, mother ID, sex, and phenotype placeholder.
- * This workflow uses FID/IID from the FAM to match 1000 Genomes genotypes to GEUVADIS individuals.

Where it enters this workflow:

- * 'nf/main.nf' resolves '\${params.plink_prefix}.bed', '.bim', and '.fam'.
- * 'FILTER_EUROPEANS' reads the '.fam' file to keep only SDRF samples that exist in the genotype data.
- * 'GENERATE_PCA' passes the prefix to GCTA through '--bfile'.
- * 'PREPARE_PHENOTYPES' also receives the '.fam' file for sample-ID consistency.

Why we use it:

- * GREML needs a GRM derived from genotype data for the same individuals whose expression is being modeled.
- * The 1000 Genomes EUR genotype resource is the matched genotype backbone for GEUVADIS European expression samples.

Critical caveat:

- * The configured prefix ends in '.22'. A successor must verify whether this is intentionally chromosome 22 only or whether production should use a genome-wide merged 1000 Genomes EUR prefix.
- * If this is chromosome 22 only, 'h2_GREML' estimates are chromosome-22-tagged variance components, not genome-wide SNP heritability.

HapMap3/no-MHC SNP list:

```
/gpfs/data/mostafavilab/shared_data/LDSC_data/hm3_no_mhc_snps.txt
```

What it contains:

- * A list of SNP IDs used with GCTA '--extract'.
- * The intended variant set is HapMap3-style common, well-imputed, widely used SNPs excluding the MHC region.

Where it enters this workflow:

- * When '--use_hm3_no_hla true', 'main.nf' validates this file and sets:

```
--extract /gpfs/data/mostafavilab/shared_data/LDSC_data/hm3_no_mhc_snps.txt
```

- * 'GENERATE_PCA' passes that extraction argument to GCTA while building the GRM.

Why we use it:

- * HapMap3 SNPs are a conservative common-SNP set often used in heritability and LDSC-style workflows.
- * Excluding the MHC avoids a region with unusually long-range LD and strong immune-related effects that can dominate covariance patterns.
- * Comparing 'all_snps' against 'hm3_no_mhc' tests how sensitive expression heritability estimates are to the SNP set used to build the GRM.

Optional unrelated keep file:

```
params.unrelated_keep_file = ""
```

Example command-line use:

```
--unrelated_keep_file /path/to/geuvadis_1000g_unrelated_eur.keep
```

Expected content:

- * Either one column containing IID, or two columns containing FID and IID.
- * The workflow accepts both formats. If two columns are present, it first tries exact FID/IID matching and then falls back to IID matching if needed.

Where it enters this workflow:

- * 'FILTER_EUROPEANS' intersects the mapped GEUVADIS EUR samples with this keep list.
- * The resulting 'eur_keep.txt' is passed to GCTA '--keep' during GRM/PCA construction.

Why we use it:

- * Hernandez-style GEUVADIS analyses remove cryptically related samples before heritability analysis.
- * The keep-list approach is different from blindly using GCTA '--grm-cutoff 0.025'. A keep list lets the user reproduce a specific unrelated panel from 1000 Genomes relationship metadata or PLINK KING/IBD pruning.

Caveat:

- * If phenotype and GRM files are reused with '--reuse_run_dir', the unrelated keep file does not change those reused artifacts. The reused files must already have been created from the same sample set.

GCTA executable:

```
/gpfs/data/mostafavilab/programs/GCTA/gcta-1.95.0-linux-kernel-3-x86_64/squashfs-root/AppRun
```

What it contains:

- * The local GCTA 1.95 AppRun executable used for GRM construction, PCA, GREML, and optionally HE regression.

Where it enters this workflow:

- * 'GENERATE_PCA' calls it with '--make-grm' and '--pca'.
- * 'ESTIMATE_HERITABILITY' calls it with '--reml' for every gene.
- * 'run_he=false' is currently the default because HE regression hit a missing MKL runtime library in the AppRun environment.

Why we use it:

- * GCTA GREML is the core variance-component estimator. It models covariance among individuals as a function of the genotype-derived GRM.
- * Using one explicit executable path avoids version drift across login nodes or modules.

Phenotype-preparation script:

```
/gpfs/data/mostafavilab/sool/analysis/GeneExpression/20260428_GE_GEUVADIS_v2/GeneExpression/nf/bin/prepare_phenotypes.R
```

What it contains:

- * The R implementation of sample matching, GTEx-style expression filtering, TMM calculation, raw/IRNT phenotype transformation, PEER factor estimation, covariate file generation, and GCTA phenotype formatting.

Where it enters this workflow:

- * 'PREPARE_PHENOTYPES' runs it through 'Rscript'.
- * 'run_greml.sh' and 'main.nf' both validate that the script contains expected diagnostic markers.
- * 'main.nf' refuses to run if the selected script hash differs from the canonical 'nf/bin' copy.

Why we use it:

- * This is where most of the biological phenotype definition happens.
- * The hash checks make the workflow safer after multiple debugging iterations because they prevent an older copy of the phenotype script from silently being used.

Conda/R runtime for phenotype preparation:

```
module load anaconda3/gpu/new
source activate r_peer
```

What it contains:

- * R and required packages such as 'data.table', 'edgeR', and 'peer'.

Where it enters this workflow:

- * The Slurm profile applies this 'beforeScript' to 'FILTER_EUROPEANS' and 'PREPARE_PHENOTYPES'.

Why we use it:

- * 'edgeR' is required for TMM normalization.
- * 'peer' is required to estimate PEER hidden factors.
- * Keeping these in a named environment improves reproducibility and avoids relying on whatever R packages happen to be installed on a login node.

Nextflow and Java runtime:

```
module load nextflow
```

What it contains:

- * Nextflow and its Java runtime dependency on the HPC module system.

Where it enters this workflow:

- * 'run_greml.sh' loads Nextflow before launching 'nf/main.nf'.

Why we use it:

- * Nextflow coordinates the many per-gene Slurm jobs, handles caching/resume, and publishes outputs into the run directories.

Run directory root:

```
/gpfs/data/mostafavilab/sool/analysis/GeneExpression/20260428_GE_GEUVDIS_v2/GeneExpression/runs
```

What it contains:

- * One subdirectory per parameter combination, for example:

```
all_snps_tmm_raw_peerauto_pmg0_npc5
hm3_no_mhc_tpm_irnt_peerauto_pmg0_npc5
```

Where it enters this workflow:

- * 'run_greml.sh' sets each Nextflow launch directory to 'runs/<combo-label>'.
- * 'params.outdir' becomes 'runs/<combo-label>/results'.

Why we use it:

- * Separate run directories prevent different SNP/expression/normalization combinations from overwriting each other.
- * Keeping logs and results together makes it much easier to debug a specific parameter combination.

Scratch/work directory root:

```
/gpfs/scratch/sl8085/nextflow_work/greml_production
```

What it contains:

- * Nextflow task work directories.
- * Per-gene '.command.sh', '.command.run', '.command.out', '.command.err', staged input files, and temporary GCTA outputs.

Where it enters this workflow:

- * 'run_greml.sh' exports 'NXF_WORK'.
- * 'nextflow -w' points to the run-specific scratch directory.

Why we use it:

- * Per-gene GREML produces many temporary files. Scratch storage is more appropriate than the lab project directory for high-volume Nextflow work.
- * When debugging, this is where the exact per-task GCTA command and GCTA log can be found.

Reusable phenotype inputs from prior runs:

```
runs/<run-name>/results/data/pheno_<snp>_<source>_<normalization>.phenotypes.tsv
runs/<run-name>/results/data/pheno_<snp>_<source>_<normalization>.qcovar
runs/<run-name>/results/data/pheno_<snp>_<source>_<normalization>.gene_index_map.txt
runs/<run-name>/results/data/pheno_<snp>_<source>_<normalization>.filtered_gene_ids.txt
```

What the phenotype file contains:

- * GCTA-compatible phenotype matrix with no header.
- * First two columns are FID and IID.
- * Each remaining column is one gene phenotype.

What the qcovar file contains:

- * GCTA-compatible quantitative covariate matrix with no header.
- * First two columns are FID and IID.
- * Remaining columns are genotype PCs and PEER factors.

What the gene index map contains:

- * Headered table linking 'gene_name' to 'mpheno_index'.
- * GCTA uses '--mpheno <index>' to choose the gene column in the phenotype matrix.

What the filtered gene list contains:

- * Genes that passed the mandatory TPM/count expression filter before phenotype transformation.

Where they enter this workflow:

- * If '--reuse_pheno_dir' or '--reuse_run_dir' is set, 'main.nf' can skip 'PREPARE_PHENOTYPES' and feed these files directly into 'ESTIMATE_HERITABILITY'.

Why we use them:

- * Preparing phenotypes can take many hours because PEER is fit over a large gene-by-sample matrix.
- * Reusing these files is appropriate when only the GCTA/GREML part changed and the sample set, expression source, normalization, PCs, and PEER settings are unchanged.

Reusable GRM/PCA inputs from prior runs:

```
runs/<run-name>/results/pca/<snp>_<source>_<normalization>/genotype_grm.grm.bin
runs/<run-name>/results/pca/<snp>_<source>_<normalization>/genotype_grm.grm.N.bin
runs/<run-name>/results/pca/<snp>_<source>_<normalization>/genotype_grm.grm.id
runs/<run-name>/results/pca/<snp>_<source>_<normalization>/geno_pca.eigenvec
```

What they contain:

- * 'grm.bin' contains the binary lower-triangle GRM values.

- * 'grm.N.bin' contains the number of SNPs used for each pairwise relationship estimate.
- * 'grm.id' contains the FID/IID order of individuals in the GRM.
- * 'geno_pca.eigenvec' contains genotype principal components for the kept samples.

Where they enter this workflow:

- * If '--reuse_grm_dir' or '--reuse_run_dir' is set, 'main.nf' can skip 'FILTER_EUROPEANS' and 'GENERATE_PCA' and stage the existing 'genotype_grm*' files into GREML tasks.

Why we use them:

- * GRM/PCA files depend only on genotype prefix, SNP set, and kept samples. They do not depend on TPM versus TMM or raw versus IRNT expression.
- * Reusing them saves time and avoids needless re-creation of identical genotype artifacts.

Downstream summary input files:

```
runs/<run-name>/results/summary/final_heritability_summary_<snps>_<source>_<normalization>.tsv
```

What they contain:

- * One row per gene.
- * Columns include 'Gene', 'Status', 'Index', 'h2_GREML', 'SE_GREML', 'Pval_GREML', 'h2_HE', and 'SE_HE'.

Where they enter downstream analyses:

- * 'scripts/plot_greml_h2_summary_boxplot.R' reads them for global h2 distributions.
- * 'scripts/pairwise_h2_comparisons.R' reads them for pairwise h2 scatter plots.
- * 'scripts/tmm_expression_h2_deciles.R' reads them for expression decile analyses.
- * 'scripts/tmm_raw_skewness_analysis.R' reads them for skewness versus h2 analyses.

Why we use them:

- * They are the canonical per-run GREML result table.
- * Plotting scripts should use these summary files rather than scratch work directories because the summary files are stable, published outputs.

Downstream phenotype inputs used by plotting/QC scripts:

```
runs/<run-name>/results/data/pheno_<snps>_<source>_<normalization>.phenotypes.tsv
runs/<run-name>/results/data/pheno_<snps>_<source>_<normalization>.gene_index_map.txt
```

What they contain:

- * The phenotype matrix contains sample-level expression phenotypes after the selected expression source, normalization mode, and PEER/PCA covariate preparation.
- * The gene index map maps phenotype columns back to Ensembl gene IDs.

Where they enter downstream analyses:

- * 'scripts/tmm_expression_h2_deciles.R' reads RAW TMM and RAW TPM phenotype files to compute expression means, medians, deciles, and skewness bins.
- * 'scripts/tmm_raw_skewness_analysis.R' reads RAW TMM phenotype files to compute skewness per gene and selected expression-distribution examples.
- * 'scripts/tpm_tmm_peer_correlations.R' reads TPM and TMM phenotype files to compare post-processing expression values gene-by-gene.
- * 'scripts/pairwise_h2_comparisons.R' reads RAW expression phenotype files to color h2 scatter plots by log2 mean expression.

Why we use them:

- * Summary files contain h2 but not the underlying expression distribution.
- * Phenotype files are needed to ask whether h2 differences are associated with expression abundance, zero inflation, skewness, or TPM/TMM phenotype disagreement.

Caveat:

* The phenotype files are already transformed according to the run setting. For expression–distribution questions, use RAW TMM or RAW TPM runs rather than IRNT runs, because IRNT intentionally erases the native expression scale.

Analysis output root on HPC:

```
/gpfs/data/mostafavilab/sool/analysis/GeneExpression/20260428_GE_GEUVDIS_v2/GeneExpression/runs/_analysis
```

What it contains:

- * Organized downstream tables and plots produced by the R scripts in 'scripts/'.
- * Subdirectories include 'plot_greml_h2_summary_boxplot', 'pairwise_h2_comparisons', 'tmm_expression_h2_deciles', 'tmm_raw_skewness', and 'tpm_tmm_peer_correlations'.

Why we use it:

- * It separates exploratory/diagnostic plots from per–run Nextflow outputs.
- * It makes it easier to copy or archive analysis products without touching Nextflow cache or scratch directories.

Slurm and Nextflow log locations:

```
runs/nextflow_logs/nextflow_driver-<combo_label>-<jobid>.out  
runs/nextflow_logs/nextflow_driver-<combo_label>-<jobid>.err  
runs/<run-name>/nextflow_logs/nextflow-<jobid>.log
```

What they contain:

- * Driver stdout/stderr from the Slurm job that launches Nextflow.
- * Nextflow engine logs for a specific run session.
- * Launch information, resolved paths, resume status, run key, work directory, parameter values, task failures, and work–directory pointers.

Why we use them:

- * These are the first place to look when a run fails before producing a final summary.
- * They provide the exact work directory for failed tasks, which can then be inspected through '.command.sh', '.command.err', '.command.out', '.exitcode', and GCTA logs.

Run–specific scratch work directories:

```
/gpfs/scratch/sl8085/nextflow_work/greml_production/<run-name>
```

What they contain:

- * Hashed Nextflow task directories for 'FILTER_EUROPEANS', 'GENERATE_PCA', 'PREPARE_PHENOTYPES', 'ESTIMATE_HERITABILITY', 'SUMMARIZE_RESULTS', and 'COMPRESS_PHENOTYPE'.
- * Per–gene intermediate GCTA files such as '<gene>_greml.gcta.log', '<gene>_greml.hsqa', '<gene>.stats', and '.command.*' wrapper files.

Why we use them:

- * Published results are clean and compact, while scratch work directories preserve the exact command–level evidence needed for debugging.
- * If a gene fails or is marked 'FAIL', the scratch task directory is where to inspect the GCTA reason.

5. Upstream RNA–seq Quantification

The previous Nextflow workflow quantified GEUVADIS RNA–seq with Salmon using GRCh37/hg19 Ensembl 75 transcriptome references:

```
Homo_sapiens.GRCh37.75.gtf.gz  
Homo_sapiens.GRCh37.75.cdna.all.fa.gz
```

It did the following:

1. Build a Salmon transcriptome index from the cDNA fasta.
2. Build a 'tx2gene.tsv' map from the GTF.
3. Run 'salmon quant' per GEUVADIS FASTQ using:

```
--validateMappings
--seqBias
--gcBias
--geneMap tx2gene.tsv
```

4. Merge per-sample 'quant.genes.sf' into:

```
gene_counts.tsv
gene_tpm.tsv
```

The Salmon workflow itself produced TPM and estimated counts. It did not create TMM. TMM is calculated later during phenotype preparation from the gene-level count matrix.

6. Main Nextflow Workflow Logic

The main workflow is in:

```
nf/main.nf
```

The main configuration is in:

```
nf/nextflow.config
```

The Slurm driver is:

```
run_greml.sh
```

Step 1: Slurm driver and fanout

The usual entry point is:

```
sbatch run_greml.sh
```

If no CLI arguments are supplied, 'run_greml.sh' defaults to fanout mode and submits 12 child jobs:

```
2 SNP sets x 2 expression sources x 3 normalization modes = 12 runs
```

Default fanout dimensions:

```
SNP sets: hm3_no_mhc, all_snps
Expression sources: tpm, tmm
Normalization types: irnt, inverse_normal, raw
```

Current child run labels look like:

```
all_snps_tmm_raw_peerauto_pmg0_npc5
all_snps_tmm_irnt_peerauto_pmg0_npc5
hm3_no_mhc_tpm_inverse_normal_peerauto_pmg0_npc5
```

The script prints a combo hash, but the current run directory name does not include that hash.

Step 2: FILTER_EUROPEANS

This process:

1. Reads GEUVADIS SDRF metadata.
2. Detects the population column by searching for 'ancestry category'.
3. Detects the RNA run column by searching for 'ENA_RUN'.
4. Keeps European groups:

```
British
Finnish
Tuscan
Utah
```

5. Maps GEUVADIS samples to PLINK '.fam' IIDs.
6. Optionally intersects with 'unrelated_keep_file'.
7. Emits:

```
eur_keep.txt
eur_map.tsv
```

'eur_keep.txt' is the FID/IID keep list used by GCTA. 'eur_map.tsv' maps RNA run IDs to genotype FID/IID and is used by phenotype preparation.

Step 3: GENERATE_PCA

This process builds the GRM:

```
gcta --bfile <plink prefix> \
  --keep eur_keep.txt \
  <optional --extract hm3_no_mhc_snps.txt> \
  --make-grm \
  --out genotype_grm \
  --thread-num <cpus>
```

Then it runs PCA from the GRM:

```
gcta --grm genotype_grm \
  --pca 5 \
  --out geno_pca \
  --thread-num <cpus>
```

Outputs are published to:

```
results/pca/<runLabel>/
```

Key outputs:

```
geno_pca.eigenvec
genotype_grm.grm.bin
genotype_grm.grm.N.bin
genotype_grm.grm.id
genotype_grm.log
```

Scientific rationale:

- * The GRM defines expected genetic covariance among individuals.
- * PCA captures broad ancestry structure and is included as a fixed covariate.
- * HM3/no-MHC reduces complexity from high-LD and difficult regions and improves comparability with common SNP reference analyses.

Step 4: PREPARE_PHENOTYPES

This process runs:

```
nf/bin/prepare_phenotypes.R
```

It receives:

```
TPM matrix
count matrix
eur_map.tsv
geno_pca.eigenvec
PLINK .fam
phenotype prefix
PEER settings
GTEx-style filter thresholds
expression source
normalization type
```

The script performs:

1. Sample mapping and ordering.
2. Gene ID cleanup.
3. Duplicate gene handling.
4. Mandatory expression filter.
5. TMM or TPM phenotype–source selection.
6. RAW or inverse–normal phenotype normalization.
7. PEER factor estimation.
8. PCA and PEER covariate merge.
9. GCTA–compatible phenotype and qcovar file writing.

Outputs are published to:

```
results/data/
```

Important files:

```
pheno_<runLabel>.phenotypes.tsv
pheno_<runLabel>.qcovar
pheno_<runLabel>.gene_index_map.txt
pheno_<runLabel>.filtered_gene_ids.txt
pheno_<runLabel>.phenotypes.tsv.gz
```

The phenotype and qcovar files intentionally have no header because GCTA expects that format. The gene index map has a header.

Step 5: ESTIMATE_HERITABILITY

This process runs once per gene. The workflow reads the gene index map, then submits one 'ESTIMATE_HERITABILITY' task per gene.

GREML command:

```
gcta --grm genotype_grm \
  --pheno pheno_<runLabel>.phenotypes.tsv \
  --mphen <gene index> \
  --qcovar pheno_<runLabel>.qcovar \
  --reml \
  --out <gene>_greml \
  --thread-num 1
```

The output '.hsq' file contains:

```
V(G)
V(e)
Vp
V(G)/Vp
Pval
```

The pipeline extracts:

```
h2_GREML
SE_GREML
Pval_GREML
```

HE regression is implemented but disabled by default:

```
run_he = false
```

This was disabled because earlier HE runs failed due an Intel MKL shared-library issue in the GCTA AppRun environment. GREML is the primary output.

Step 6: SUMMARIZE_RESULTS

This process concatenates one '.stats' file per gene into:

```
results/summary/final_heritability_summary_<runLabel>.tsv
```

Columns:

```
Gene  
Status  
Index  
h2_GREML  
SE_GREML  
Pval_GREML  
h2_HE  
SE_HE
```

Each gene gets 'PASS' or 'FAIL'. Per-gene failures do not necessarily fail the whole workflow because the workflow writes a '.stats' row and exits the gene job cleanly.

7. Parameter Table And Scientific Meaning

Sample and genotype parameters

```
eur_pops = ["British", "Finnish", "Tuscan", "Utah"]
```

Meaning:

- * Keeps the four European GEUVADIS populations.
- * Excludes YRI because the primary analysis is EUR-only.
- * Mimics GEUVADIS/Hernandez-style European analysis.

```
unrelated_keep_file = ""
```

Meaning:

- * Optional one-column IID or two-column FID/IID file.
- * When supplied, it restricts mapped GEUVADIS EUR samples to an external unrelated sample set.
- * This is the preferred way to mimic Hernandez-style cryptic-relatedness pruning.

```
use_hm3_no_hla = true
```

Meaning:

- * Uses HapMap3 SNPs excluding the MHC/HLA region when true.
- * Rationale: HM3 SNPs are a common well-imputed/common-variant reference set; excluding MHC reduces extreme LD and complex-region behavior.

```
n_pcs = 5
```

Meaning:

- * Uses the top five genotype PCs as covariates.
- * Rationale: genotype PCs capture broad ancestry structure even within EUR.
- * This is slightly different from GTEx V7, which listed top three genotype PCs,

but five PCs is a conservative choice for a smaller population-genetic analysis.

Expression filtering parameters

```
gtex_tpm_threshold = 0.1
gtex_count_threshold = 6
gtex_sample_frac_threshold = 0.2
```

Meaning:

- * A gene must have TPM ≥ 0.1 in at least 20% of samples.
- * The same gene must also have count ≥ 6 in at least 20% of samples.
- * Both conditions must be true.

Scientific rationale:

- * TPM threshold removes near-background transcription.
- * Count threshold removes genes without enough read evidence.
- * The 20% rule prevents a gene expressed in only a few outlier individuals from entering the GREML analysis.
- * This follows GTEx-style eQTL expression filtering.

Expression-source parameters

```
expression_source = "tmm"
```

Allowed:

```
tmm
tpm
```

If 'tmm', the pipeline starts from counts, calculates TMM factors with edgeR, and uses TMM-normalized CPM.

If 'tpm', the pipeline uses Salmon-derived TPM directly.

Scientific difference:

- * TPM corrects within sample for gene/transcript length and sequencing depth.
- * TMM uses raw count composition across samples to estimate effective library size scaling factors.
- * TPM is useful for abundance; TMM is often preferred for cross-sample count comparisons because it addresses RNA composition bias.

Normalization parameters

```
normalization_type = "irnt"
```

Allowed:

```
raw
irnt
inverse_normal
```

'raw':

- * Uses TMM CPM or TPM values directly.
- * Most biologically interpretable scale.
- * More vulnerable to zero inflation and skewness.

'irnt':

- * Uses 'log2(x + 1)'.
- * Applies per-gene inverse-normal transform with Blom offset '3/8'.
- * Makes each gene approximately standard-normal across samples.
- * Less interpretable in expression units but better aligned with linear-model assumptions.

'inverse_normal':

- * Similar to 'irnt', but uses rank offset '0.5'.
- * Kept mainly for comparison.

PEER parameters

```
peer_nk = "auto"
peer_max_genes = 0
```

GTEx-style auto rule:

```
N < 150: 15 PEER factors
150 <= N < 250: 30 PEER factors
250 <= N < 350: 45 PEER factors
N >= 350: 60 PEER factors
```

Scientific rationale:

- * PEER estimates hidden factors that capture technical and biological unwanted variation, such as batch effects or broad latent expression patterns.
- * GTEx chose the sample-size rule based on optimizing eGene discovery.
- * 'peer_max_genes = 0' means use all filtered genes for PEER.
- * Setting 'peer_max_genes > 0' uses top-variance genes only and is a speed option.

GCTA parameters

```
run_he = false
```

Meaning:

- * Only GREML is run by default.
- * Haseman-Elston regression is optional and currently skipped.

```
gcta_path = /gpfs/data/mostafavilab/programs/GCTA/...
```

Meaning:

- * Exact GCTA executable used by the workflow.

8. How To Run The Workflow

Go to the project root on HPC:

```
cd /gpfs/data/mostafavilab/sool/analysis/GeneExpression/20260428_GE_GEUVDIS_v2/GeneExpression
```

Submit the default 12-run fanout:

```
sbatch run_greml.sh
```

Submit a single run:

```
GREML_FANOUT=0 sbatch run_greml.sh \
  --use_hm3_no_hla false \
  --expression_source tmm \
  --normalization_type raw
```

Run with Hernandez-style unrelated keep list:

```
GREML_FANOUT=0 sbatch run_greml.sh \  
--use_hm3_no_hla false \  
--expression_source tmm \  
--normalization_type raw \  
--unrelated_keep_file /path/to/geuvadis_1000g_unrelated_eur.keep
```

Submit 12 fanout runs with unrelated keep file:

```
GREML_FANOUT=1 GREML_FANOUT_INTERVAL_SEC=0 sbatch run_greml.sh \  
--unrelated_keep_file /path/to/geuvadis_1000g_unrelated_eur.keep
```

Disable resume:

```
GREML_RESUME=0 sbatch run_greml.sh ...
```

Use a different run root:

```
GREML_RUN_DIR=/path/to/output_root sbatch run_greml.sh ...
```

Use a different scratch root:

```
NXF_WORK_ROOT=/gpfs/scratch/<user>/nextflow_work/greml_production sbatch run_greml.sh ...
```

9. Expected Run Directory Layout

A run directory looks like:

```
runs/all_snps_tmm_raw_peerauto_pmg0_npc5/  
  nextflow_logs/  
  results/  
    data/  
    pca/  
    summary/
```

Data folder:

```
results/data/pheno_all_snps_tmm_raw.phenotypes.tsv  
results/data/pheno_all_snps_tmm_raw.phenotypes.tsv.gz  
results/data/pheno_all_snps_tmm_raw.qcovar  
results/data/pheno_all_snps_tmm_raw.gene_index_map.txt  
results/data/pheno_all_snps_tmm_raw.filtered_gene_ids.txt
```

PCA/GRM folder:

```
results/pca/all_snps_tmm_raw/genotype_grm.grm.bin  
results/pca/all_snps_tmm_raw/genotype_grm.grm.N.bin  
results/pca/all_snps_tmm_raw/genotype_grm.grm.id  
results/pca/all_snps_tmm_raw/geno_pca.eigenvec
```

Summary folder:

```
results/summary/final_heritability_summary_all_snps_tmm_raw.tsv
```

10. Quality Checks After Running

Check final summary exists:

```
ls runs/*_peerauto_pmg0_npc5/results/summary/final_heritability_summary*.tsv
```

Check PASS/FAIL counts:

```
awk -F'\t' 'NR>1 {n[$2]++} END{for (k in n) print k,n[k]}' \
runs/all_snps_tmm_raw_peerauto_pmg0_npc5/results/summary/final_heritability_summary*.tsv
```

Check sample count in GRM:

```
wc -l runs/all_snps_tmm_raw_peerauto_pmg0_npc5/results/pca/all_snps_tmm_raw/genotype_grm.grm.id
```

Check samples against PLINK 'fam':

```
PLINK_PREFIX=/gpfs/data/mostafavilab/shared_data/LDSC_data/1000G_EUR_Phase3_plink/1000G.EUR.QC.22
RUN_DIR=/gpfs/data/mostafavilab/sool/analysis/GeneExpression/20260428_GE_GEUADIS_v2/GeneExpression/runs
/all_snps_tmm_raw_peerauto_pmg0_npc5
comm -23 \
<(awk '{print $2}' "$RUN_DIR/results/pca/all_snps_tmm_raw/genotype_grm.grm.id" | sort -u) \
<(awk '{print $2}' "${PLINK_PREFIX}.fam" | sort -u)
```

No output means all used samples are in the genotype panel.

11. Downstream Analysis Scripts And Plots

All diagnostic analyses write under:

```
runs/_analysis/
```

Global h2 summary boxplot

Script:

```
scripts/plot_greml_h2_summary_boxplot.R
```

Outputs:

```
runs/_analysis/plot_greml_h2_summary_boxplot/tables/summary_of_summaries.tsv
runs/_analysis/plot_greml_h2_summary_boxplot/plots/greml_h2_boxplot_by_setting.png
```

Scientific rationale:

- * Shows distribution of PASS 'h2_GREML' estimates across all settings.
- * Helps identify which preprocessing choices inflate, deflate, or destabilize h2.
- * Mean labels are useful because h2 distributions can be skewed.

Pairwise h2 comparisons

Script:

```
scripts/pairwise_h2_comparisons.R
```

Main plot categories:

```
SNP set: ALL vs HM3/no-MHC
Expression source: TMM vs TPM
Normalization: RAW vs IRNT
```

Scientific rationale:

- * Each scatter plot compares gene-by-gene h2 estimates between two settings.
- * Identity line shows perfect agreement.
- * Color by expression level tests whether low expression drives disagreement.
- * Color by Wald Z tests whether discrepancies are concentrated among weak or strong h2 estimates.

* Discrepancy counts quantify genes where one method gives high h2 and another gives near-zero h2.

Expression decile h2 analysis

Script:

```
scripts/tmm_expression_h2_deciles.R
```

Outputs:

```
runs/_analysis/tmm_expression_h2_deciles/plots/tmm_expression_h2_deciles_boxplot.png  
runs/_analysis/tmm_expression_h2_deciles/plots/median_tpm_expression_h2_deciles_boxplot.png  
runs/_analysis/tmm_expression_h2_deciles/plots/expression_skewness_by_decile_boxplot.png
```

Scientific rationale:

- * Tests whether gene-level h2 is higher for more highly expressed genes.
- * Uses TMM RAW mean expression and TPM RAW median expression as abundance bins.
- * Filters very low expression genes to avoid bins dominated by near-zero signals.
- * Adds skewness by expression bin to check whether distribution shape changes across abundance strata.

Default thresholds:

```
MIN_LOG2_MEAN_TMM_PLUS1 = 1.5  
MIN_LOG2_MEDIAN_TPM_PLUS1 = 0
```

TMM raw skewness analysis

Script:

```
scripts/tmm_raw_skewness_analysis.R
```

Outputs include:

```
tmm_raw_skewness_hist_density_<METHOD>.png  
tmm_raw_skewness_mean_vs_skew_<METHOD>.png  
tmm_raw_skewness_selected_gene_distributions_<METHOD>.png  
tmm_vs_tpm_h2_discrepancy_expr_distributions_<METHOD>.png  
all_raw_h2_vs_hm3_raw_h2_tmm_colored_by_log2meanexpr_cap*.png  
all_raw_h2_vs_hm3_raw_h2_tmm_colored_by_skewness_cap*.png
```

Scientific rationale:

- * Skewness detects whether raw expression phenotypes are dominated by long right tails or zero inflation.
- * Selected gene histograms give concrete examples for lab discussion.
- * H2 discrepancy examples test whether TPM/TMM disagreement is visible at the expression distribution level.
- * ALL-vs-HM3 colored scatter tests whether SNP-set sensitivity is linked to expression abundance or skewness.

TPM vs TMM phenotype correlations

Script:

```
scripts/tpm_tmm_peer_correlations.R
```

Outputs:

```
tpm_tmm_peer_gene_correlations.tsv
tpm_tmm_peer_correlation_summary.tsv
tpm_tmm_peer_correlation_boxplot.png
tpm_tmm_peer_mean_scatter.png
```

Scientific rationale:

- * Compares TPM-derived and TMM-derived phenotype values directly.
- * High per-gene correlation means expression source choice is unlikely to change that gene's phenotype ranking across individuals.
- * Low correlation identifies genes where TPM/TMM choice may alter h2.

Regulatory/repeat downstream analysis

Script:

```
scripts/downstream_h2_regulatory_repeat_analysis.R
```

This is a separate biological follow-up pipeline. It links h2 to:

- * 's_het post_mean'
- * gene regulatory burden
- * enhancer/open chromatin overlap
- * repeat classes/families/subtypes
- * randomized repeat background

Scientific rationale:

- * Tests whether gene expression heritability varies with evolutionary constraint.
- * Tests whether regulatory architecture or repeat burden differs across gene bins.
- * Observed-vs-background repeat analysis asks whether repeat patterns exceed random genomic expectation.

12. Interpretation Guide For Main Plots

h2 boxplot by setting

Use this first. It answers:

- * Which pipeline choices produce higher h2?
- * Are distributions broad, compressed, or outlier-heavy?
- * Are PASS counts similar across methods?

Do not interpret differences without checking PASS/FAIL counts.

Pairwise h2 scatter plots

Use these to diagnose gene-level stability.

Interpretation:

- * Points on diagonal: stable gene h2 across methods.
- * Points above diagonal: y-axis method estimates higher h2.
- * Points below diagonal: x-axis method estimates higher h2.
- * Low-expression colored outliers: likely expression-scale or low-information issue.
- * High-Z outliers: more likely biologically meaningful or at least statistically stable.

Expression decile h2 plots

Use these to ask whether expression abundance drives h^2 .

Interpretation:

- * Increasing h^2 across deciles: low-expression genes may suppress h^2 estimates.
- * Flat h^2 across deciles: h^2 estimates are robust to abundance after filtering.
- * Mean greater than median: h^2 distribution is right-skewed in that bin.

Skewness plots

Use these to ask whether raw expression distributions are suitable for GREML.

Interpretation:

- * High positive skew: a few individuals have much higher expression.
- * High zero fraction: raw-scale h^2 may be unstable.
- * IRNT should reduce sensitivity to skew because it rank-normalizes per gene.

TPM/TMM correlation plots

Use these to ask whether expression source changes phenotype rankings.

Interpretation:

- * High correlation: TPM and TMM mostly preserve individual-level ordering.
- * Low correlation: TPM/TMM choice can change the phenotype and therefore h^2 .

13. Known Caveats

1. The default genotype prefix ends with '.22'. Confirm whether the analysis is intentionally chromosome 22 only or whether genome-wide genotype files should be merged or supplied.
2. 'params.grm_prefix' exists but the code currently uses hard-coded 'genotype_grm'.
3. 'run_he=true' may fail unless the GCTA AppRun MKL runtime issue is fixed. The current workflow intentionally sets 'run_he=false'.
4. Full reuse mode skips sample filtering and phenotype preparation. If 'unrelated_keep_file' is set while reusing GRM and phenotype files, the keep file does not change those reused artifacts.
5. Phenotype-only reuse with a newly generated GRM can mismatch normalization and sample set. The workflow warns but does not stop.
6. Per-gene GREML failures do not necessarily fail the whole workflow. Always inspect 'Status' in the summary table.
7. Several downstream scripts have hard-coded HPC paths. Before rerunning on a new project copy, inspect the first 20–40 lines of each R script.
8. Plot files are generated on the HPC run directory under 'runs/_analysis/'; they are not stored in this local repo by default.

14. Minimal Handoff Checklist

Before a new analyst reruns the workflow:

1. Confirm the genotype prefix is correct and genome-wide if needed.

2. Confirm the GEUVADIS TPM/count matrices exist.
3. Confirm 'gcta_path' works on compute nodes.
4. Confirm the 'r_peer' environment has 'data.table', 'edgeR', 'peer', and ideally 'matrixStats'.
5. Decide whether to provide 'unrelated_keep_file'.
6. Decide whether to run all 12 settings or a single setting.
7. Keep 'run_he=false' unless the MKL issue is solved.
8. Run a single small/default job first if changing input paths.
9. Check sample count in 'genotype_grm.grm.id'.
10. Check PASS/FAIL counts in the final summary.
11. Run diagnostic plots under 'scripts/'.
12. Archive both 'nextflow_logs/' and 'results/'.

15. Source List

GEUVADIS/1000G:

* International Genome Sample Resource GEUVADIS data collection:

<https://www.internationalgenome.org/data-portal/data-collections/geuvadis.html>

* GEUVADIS project summary: RNA-seq from 1000 Genomes samples across CEU, FIN, GBR, TSI, and YRI.

Salmon:

* Patro et al. Salmon provides fast and bias-aware quantification of transcript expression. Nature Methods 2017. <https://www.nature.com/articles/nmeth.4197>

TMM:

* Robinson and Oshlack. A scaling normalization method for differential expression analysis of RNA-seq data. Genome Biology 2010. <https://genomebiology.biomedcentral.com/articles/10.1186/gb-2010-11-3-r25>

GTEx methods:

* GTEx Analysis Methods V7 PDF.

https://storage.googleapis.com/gtex-public-data/Portal_Analysis_Methods_v7_09052017.pdf

GCTA/GREML:

* Yang et al. GCTA: A Tool for Genome-wide Complex Trait Analysis. American Journal of Human Genetics 2011. <https://www.sciencedirect.com/science/article/pii/S0002929710005987>

* GCTA documentation and notes on GRM and relatedness pruning:

<https://yanglab.westlake.edu.cn/software/gcta/>

GEUVADIS expression heritability:

* Hernandez et al. Ultrarare variants drive substantial cis heritability of human gene expression. Nature Genetics 2019. <https://www.nature.com/articles/s41588-019-0487-7>

* Dahl et al. On Negative Heritability and Negative Estimates of Heritability. Genetics 2020. <https://academic.oup.com/genetics/article/215/2/343/5930411>